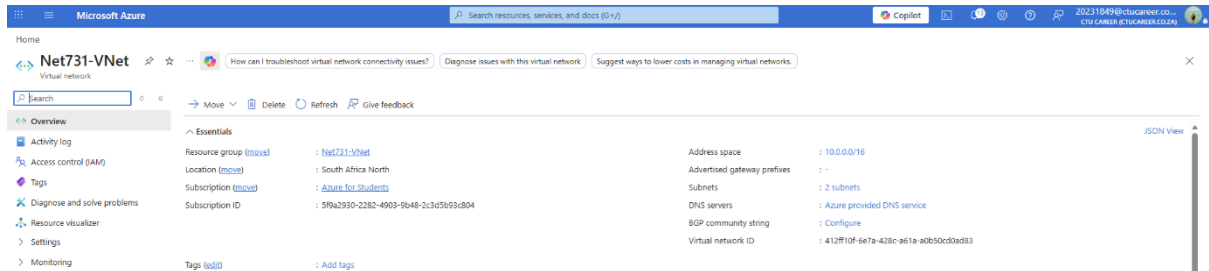


Jayden Hamilton
20231849
Week 2 Pracs

The address space 10.0.0.0/16 was selected because it falls within the RFC 1918 private IP range, provides 65,534 usable host addresses, and gives TechBridge Solutions room to grow without redesigning the network. The /16 mask allows flexible subnet division for both public-facing and internal systems.



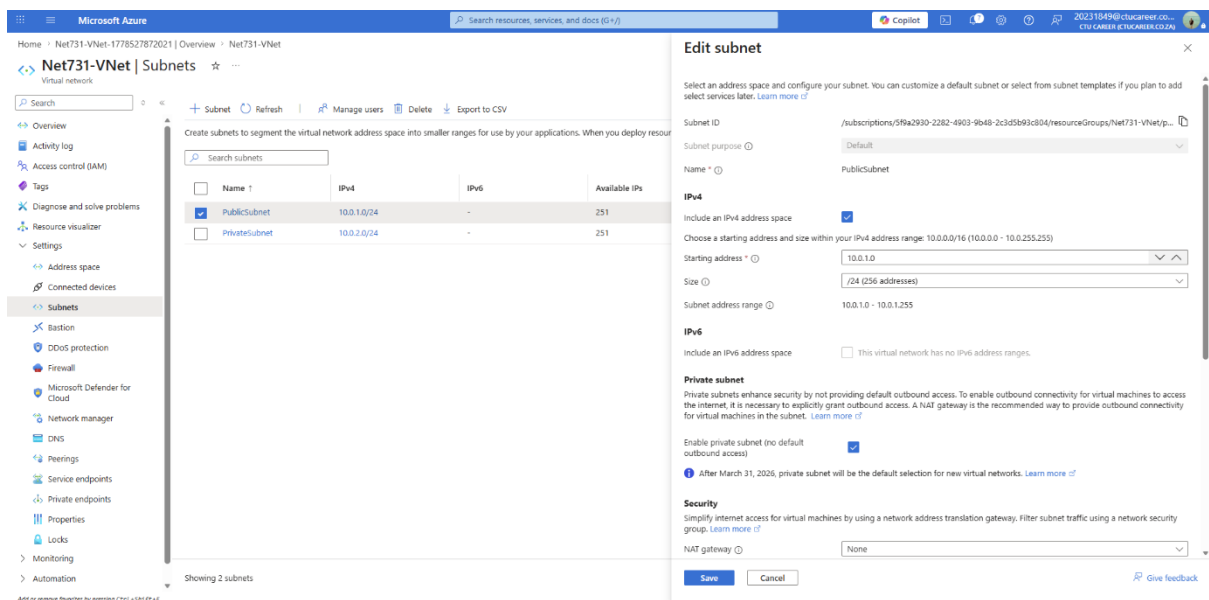
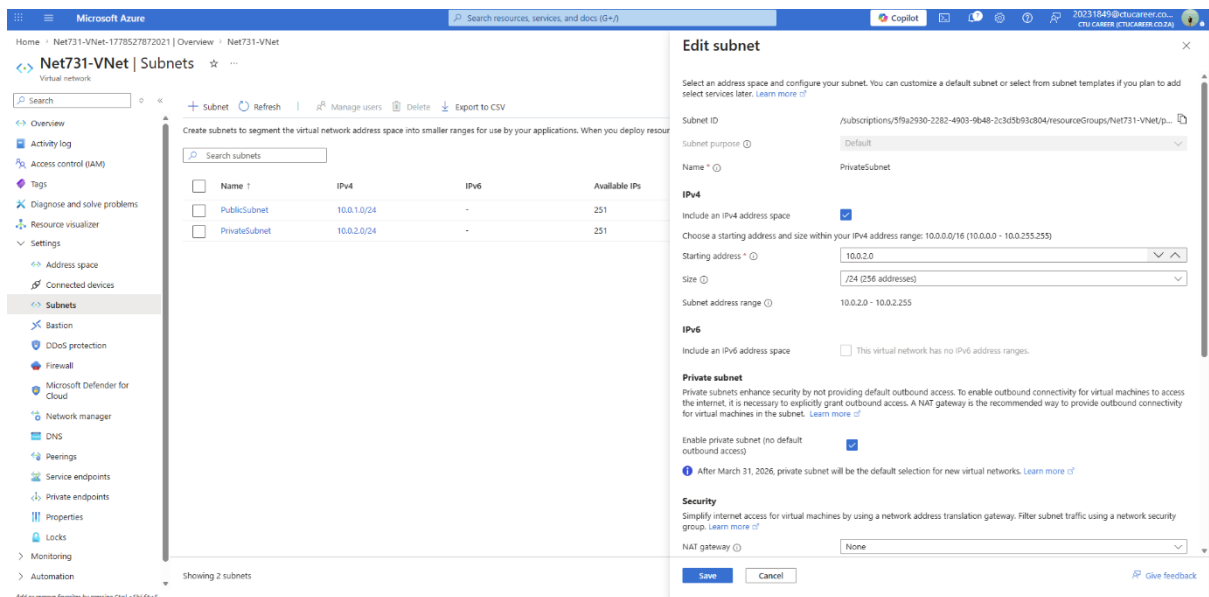
Subnet Design Justification

PublicSubnet10.0.1.0/24 Public-facing resources (e.g. web servers) This subnet hosts resources that need to be accessible from the internet. A /24 mask provides 251 usable addresses, which is sufficient for a small business like TechBridge while leaving room for growth. It is intentionally separated from internal systems to limit exposure in case of a breach.

PrivateSubnet10.0.2.0/24 Internal back-end systems (e.g. databases) This subnet hosts sensitive internal resources that should never be directly accessible from the internet. Isolating it from the PublicSubnet ensures that even if the public-facing layer is compromised, the back-end systems remain protected. A /24 mask matches the public subnet for consistency and scalability.

Naming Convention Justification

Subnet names were chosen to clearly reflect their role in the network. PublicSubnet and PrivateSubnet immediately communicate the security boundary each segment represents, making the network easier to manage and audit.



Allow-HTTP: Permits standard web traffic to reach public-facing resources on port 80

Allow-HTTPS: Permits encrypted web traffic on port 443, ensuring secure communication with users

Deny-RDP-Internet: Blocks Remote Desktop access from the internet on port 3389 to prevent unauthorised remote access to virtual machines

Allow-HTTPTCP80InboundAllow Permits standard HTTP traffic from the internet to reach the public-facing web servers in the PublicSubnet. Without this rule, users would not be able to access the business website.

Allow-HTTPSTCP443InboundAllowPermits encrypted HTTPS traffic to the PublicSubnet, ensuring all sensitive user data is transmitted securely. This is essential for any modern business handling customer information.

Deny-RDP-InternetTCP3389InboundDenyBlocks Remote Desktop Protocol access from the internet to prevent unauthorised users from remotely connecting to virtual machines. RDP is a common attack vector and should never be openly exposed.

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Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name

Port == all

Protocol == all

Source == all

Destination == all

Action == all

Priority	Name	Port	Protocol	Source
100	Allow-HTTP	80	TCP	Any
110	Allow-HTTPS	443	TCP	Any
200	Deny-RDP-Internet	3389	TCP	Any
65000	AllowVnetInBound	Any	Any	VirtualNetwork
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer
65500	DenyAllInBound	Any	Any	Any

Allow-HTTP

Net731-VNet

Source

Any

Source port ranges

*

Destination

Any

Service

HTTP

Destination port ranges

80

Protocol

☐ Any

☒ TCP

☐ UDP

☐ ICMPv4

☐ ICMPv6

Action

☒ Allow

☐ Deny

Priority

100

Name

Allow-HTTP

Description

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200	Deny-RDP-Internet	3389	TCP	Any
65000	AllowVnetInBound	Any	Any	VirtualNetwork
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer
65500	DenyAllInBound	Any	Any	Any

Allow-HTTPS

Net731-VNet

Source

Any

Source port ranges

*

Destination

Any

Service

HTTPS

Destination port ranges

443

Protocol

☐ Any

☒ TCP

☐ UDP

☐ ICMPv4

☐ ICMPv6

Action

☒ Allow

☐ Deny

Priority

110

Name

Allow-HTTPS

Description

Save

Cancel

Give feedback

https://portal.azure.com/#@ctucareer.co.za/resource/subscriptions/5f9a2930-2282-4903-9648-2c3d5931304/resourcegroups/Net731-VNet/providers/Microsoft.Network/networkSecurityGroups/Net731-VNet/inboundSecurityRules

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Filter by name

Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-HTTP	80	TCP	Any	Any	Allow
110	Allow-HTTPS	443	TCP	Any	Any	Allow
200	Deny-RDP-Internet	3389	TCP	Any	Any	Deny
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	AzureLoadBalancer	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Deny-RDP-Internet

Net731-VNet

Source: Any

Source port ranges: *

Destination: Any

Service: RDP

Destination port ranges: 3389

Protocol: TCP

Action: Deny

Priority: 200

Name: Deny-RDP-Internet

Description:

Save Cancel Give feedback

Add or remove priorities by pressing Ctrl + Shift + F

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Search subnets

Name	Address range	Virtual network
PrivateSubnet	10.0.2.0/24	Net731-VNet
PublicSubnet	10.0.1.0/24	Net731-VNet

Saving subnet
Successfully saved network security group for subnet "PrivateSubnet".

Give feedback